

BrightSign NPU Platform

Next-generation edge AI architecture for BrightSign digital signage players

This repository contains the NPU-accelerated software platform powering BrightSign's latest products based on the Rockchip RK3588 processor. The platform leverages the RK3588's integrated 3-core NPU (6 TOPS) to deliver real-time computer vision and AI analytics at the edge with no cloud dependency, zero latency, and complete data privacy.

Platform Highlights

Hardware Foundation:

The platform supports three Rockchip NPU-enabled SoCs, each targeting different performance and cost requirements:

- **RK3588** (3-core NPU, 6 TOPS) - Premium tier, used in **XT5**
- **RK3576** (2-core NPU, 4 TOPS) - Mid-tier, used in **XS156**
- **RK3568** (1-core NPU, 1 TOPS) - Entry tier, used in **LS5/HS5**

Performance characteristics (using XT5/RK3588 as reference):

- **5.5W total power** for 3-model AI workload
- **Fanless operation** with passive cooling
- **100,000-hour MTBF** with 5-year warranty - enterprise-grade reliability exceeding competitors by 3.6-4.5x

Note: This documentation primarily focuses on the XT5/RK3588 implementation. Lower-tier products (XS156, LS5/HS5) support fewer simultaneous models but use the same software architecture and C++ implementation.

Software Architecture:

- **C++ implementation** with optimized RKNN runtime for maximum efficiency
- **Multi-model parallelism** - run up to 3 AI models simultaneously on separate NPU cores
- **Zero-copy data pipelines** for minimal memory bandwidth usage
- **Sub-15ms inference latency** per model at full resolution

Performance Advantages:

- **3.3x more power-efficient** than Python implementations on identical hardware
- **1.7-2.2x lower power consumption** than competing edge AI platforms
- **Memory-efficient design** enables true 3-model parallel execution (impossible with Python on same hardware)
- **0.42 TOPS/Watt** computational efficiency vs. 0.17-0.28 for competitors

BrightShopper: Flagship Application

BrightShopper is the first product built on this NPU platform, transforming BrightSign digital signage players into intelligent shopper analytics sensors.

What BrightShopper Does

Real-time retail analytics running entirely on-device:

- **People counting & tracking** - Know exactly who's in view and where they move
- **Gaze detection & attention tracking** - Measure which content captures attention and for how long
- **Pose estimation** - 17-keypoint skeleton tracking for behavior recognition
- **Shopping behavior detection** - Cart pushing, basket carrying, shelf interaction, product selection
- **Content attribution** - Link engagement metrics to specific media (BSN.cloud Analytics)

Unique Selling Propositions

1. **Edge-first, privacy-first:** All processing on-device. No video transmitted to cloud. Zero PII collected.
2. **Real-time performance:** Millisecond-latency analytics enable immediate content adaptation and live dashboards.
3. **Unmatched efficiency:** 5.5W for full 3-model analytics 42% less power than competitors. Enables fanless 24/7 operation.
4. **Proven reliability:** Built on BrightSign's media player platform deployed in 100,000+ locations worldwide. 100,000-hour MTBF with 5-year warranty delivers 3.6-4.5x better reliability than fan-cooled alternatives.
5. **Multi-model intelligence:** Combines face detection (RetinaFace), pose estimation (YOLOv8-pose), and object detection (YOLOx) running in parallel for comprehensive behavioral insights.
6. **Content-to-engagement attribution:** BSN.cloud Analytics automatically links viewer metrics to specific media assets, proving campaign ROI.

Platform Configurations

Player	SoC	NPU Cores	Models Supported	Features
XT5	RK3588	3-core (6 TOPS)	3 models parallel	Face + Pose + Object detection - full behavioral analytics
XS156	RK3576	2-core (4 TOPS)	2 models parallel	Face + Object detection - engagement analytics
LS5/HS5	RK3568	1-core (1 TOPS)	1 model	Face detection only - basic engagement metrics

Power consumption:

- **XT5:** ~5.5W (3-core workload)
- **XS156:** ~4W estimated (2-core workload)
- **LS5/HS5:** ~3.5W estimated (1-core workload)

All configurations use the same optimized C++ implementation and share the core software architecture. Lower-tier products simply run fewer models in parallel based on available NPU cores.

Documentation

- [BrightShopper PR-FAQ](#) - Product vision, capabilities, and frequently asked questions
- [Power Analysis](#) - Detailed efficiency, thermal, and reliability analysis
- [Single-Core Power Comparison](#) - XT5 vs. OrangePi vs. Raspberry Pi benchmarks

- [Multi-Model Architecture](#) - Technical design for parallel NPU execution
- [Design Overview](#) - System architecture and implementation details
- [RGB-D Support](#) - Depth sensing capabilities

Key Metrics

Metric	XT5 Value	Industry Context
TOPS/Watt	0.42	1.5-2.3x better than competitors
R-TOPS/Watt	0.84	7-8x better lifetime computational value
Power (3-core)	5.5W	42-54% less than competitors
MTBF	100,000 hrs	3.6-4.5x better reliability than fan-cooled platforms
Incremental power/core	+0.80W	3.3x better than Python on same hardware

R-TOPS/W (Reliable-TOPS/Watt) = a novel metric combining computational efficiency (TOPS/W) with operational lifetime (MTBF), measuring the total computational value delivered per watt over the device's lifespan.

Competitive Advantages

vs. Cloud-based Analytics

- **Zero latency:** Milliseconds vs. seconds for cloud round-trip
- **No bandwidth costs:** All processing local, no video upload
- **100% uptime:** Works without internet connectivity
- **Privacy compliance:** No PII leaves the device

vs. Python Edge AI Implementations

- **3.3x lower power per NPU core:** C++ vs. Python efficiency
- **Memory bandwidth:** Python would saturate NPU bandwidth at 3 models; C++ enables true parallelism
- **Production reliability:** Compiled binary vs. interpreter dependencies

vs. External NPU Accelerators (Hailo, Coral)

- **1.8x lower total power:** No USB/PCIe overhead, native NPU integration
- **Lower baseline power:** 3.1W vs. 4.7W static
- **Simpler design:** No external modules, PoE-powered single-box solution

vs. Retail Analytics Cameras

- **Dual-purpose hardware:** Digital signage + analytics in one device
- **Lower TCO:** No separate camera infrastructure
- **Content attribution:** Link viewer data to displayed media automatically

Getting Started

Documentation for deploying BrightShopper and developing custom NPU applications coming soon.

License

License information to be determined.

BrightSign - Market leader in digital signage media players **Platform:** Rockchip RK3588 NPU-accelerated edge AI **Applications:** Retail analytics, audience measurement, behavioral intelligence